RLVR Moves Along a Coverage Frontier Set by the Base Model: Pre-Registered Evidence from RL Interventions, Warm Starts and Public Checkpoints

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Abstract

Many methods claim to make reinforcement learning with verifiable rewards (RLVR) more diverse, usually by reporting higher pass@k at a fixed number of training steps. A method that learns more slowly is less sharpened at any fixed step, so such comparisons mix speed with coverage. We compare pass@16 at matched pass@1, with decision rules registered before each run and problem-bootstrap intervals. On Qwen2.5-Math-1.5B and 7B, the interpretable comparisons of more than twenty RL-side interventions—advantage rules, two outcome-diversity bonuses, sharpness-aware updates, replay, larger groups, rollout allocation, guidance prefixes and 1,000-step training—are consistent with movement along Dr. GRPO's pass@16-versus-pass@1 curve, not with a change of the curve, and 26 public checkpoints from other groups fall on or below a curve predicted from the base model. Small supervised warm starts before RL are the only lever we found that lifts the curve: teacher solutions raise matched pass@16 by +2.1 points over early RL, beat the base model's own solutions on identical prompts (+1.53, 95% [+0.34, +2.76]), and keep their effect with corrupted final answers; coverage weighting, TailSFT-style filtering, hard-only selection and interleaving the same data during RL do not add to this. Against 700-step Dr. GRPO references, however, every start's advantage decays to about +0.4 points by step 150 (three seeds, descriptive): starts accelerate progress along the frontier and make a transient excursion above it. Per-problem decompositions locate teacher gains on problems the base model never solves. We conclude that, in these settings, RLVR's coverage at a given accuracy is a property of the starting model, and we outline what a method would have to do to change it.

1. Introduction

RLVR trains a language model on automatically checked answers (Shao et al., 2024; Guo et al., 2025). Because RL sharpens the policy, RL-trained models are often beaten by their own base model at large k (Yue et al., 2025), and a growing literature tries to keep rollouts diverse: modified advantages (Zhu et al., 2025; Walder & Karkhanis, 2025; Chen et al., 2025), exploration bonuses (Song et al., 2025), entropy control (Cui et al., 2025), rollout budgets (Hu et al., 2025; Yang et al., 2025; Li et al., 2025b; Yu et al., 2026), off-policy guidance (Yan et al., 2025; Ma et al., 2025; Li et al., 2025a; Zhang et al., 2025b) and warm starts (Dong et al., 2026; Malladi et al., 2026). Most report pass@k at a fixed step.

We separate speed from coverage by comparing a method's pass@16 with a reference curve at the same pass@1 (Section 2). Every decision rule is registered before its run; intervals resample test problems and recompute the reference in each draw. Our findings:

1. RL-side interventions stay on one curve (Section 3): more than twenty registered interventions, including the outcome-based exploration bonuses of Song et al. (2025) and sharpness-aware Flat-RLVR (Kim et al., 2026), reach average equivalence with Dr. GRPO's curve or remain within ±1.2 points; 1,000-step training raises pass@1 by 9–13 points without meeting the registered coverage criterion.
2. Public checkpoints lie on or below the base-predicted curve (Section 4): 26 checkpoints, 38 evaluations, largest residual +0.9 points.

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3. Warm starts lift the curve during early RL, and teacher content is what matters (Section 5): teacher-solution starts meet the registered curve-raised criterion, beat the base model's own solutions on identical, length-matched prompts, and survive corrupted final answers; how the SFT data is weighted, filtered or timed does not matter.

4. The excursion is transient (Section 6): against 700-step references, all start advantages decay by RL step 150, while teacher starts reach the 700-step reference's accuracy in 150 steps.

5. Where gains sit (Section 7): matched-accuracy coverage gains are across-problem spread (pass@2), and teacher gains concentrate on problems with zero base successes; self-generated starts gain on problems the base already solves sometimes, where RL catches up.

2. Setup and protocol

For problem x with binary reward, p_x is the policy's success probability. Over N problems, $P_k = \frac{1}{N}\sum_x[1-(1-p_x)^k]$, estimated without bias from n_x samples with C_x successes (Chen et al., 2021).

Proposition 2.1 (pass@k at matched pass@1 measures spread). $P_1 = \mathbb{E}_x p_x$ and $P_k = \mathbb{E}_x f_k(p_x)$ with $f_k(p) = 1-(1-p)^k$ strictly concave for $k > 1$. At fixed mean, a mean-preserving spread of $\{p_x\}$ lowers P_k; in particular $P_2 = 2P_1 - \mathbb{E}_x p_x^2$, so at matched pass@1 the pass@2 gap equals minus the change in the second moment of success across problems. For a small change δ_x with $\mathbb{E}\delta_x = 0$, the first-order pass@k change is $\mathrm{Cov}_x\big(k(1-p_x)^{k-1}, \delta_x\big)$.

Training. Dr. GRPO (Liu et al., 2025): group-relative advantages $r_i - \bar{r}$, sequence-sum loss with a constant normalizer, one update per batch of fresh rollouts, no KL term, AdamW at 10^{-6}, 16 problems × 8 rollouts per step, 150 steps, MATH training problems (Hendrycks et al., 2021). Qwen2.5-Math-1.5B (Yang et al., 2024) at 512 or 1,024 response tokens; 7B at 2,048 tokens. Reference runs (B1k, three seeds) used one L40S each; warm-start arms used A100s (Section 10).

Evaluation. 511 held-out MATH test problems, 32 samples at $T = 1$ (64 for starting models, 256 where stated), vLLM with per-problem seeds and the training prompt.

Matched-accuracy estimator. For a checkpoint with pass@1 a, the gap is $\widehat{P}_{16} - R_s(a)$, where R_s is the same-seed reference trajectory (base and saved steps, sorted by pass@1) linearly interpolated at a. Eligible checkpoints have a inside R_s's range on the full data; a seed needs two. Gaps average eligible checkpoints within a seed, then seeds equally. 2,000 problem-bootstrap draws recompute the reference; from Pre-registration 36 on, the checkpoint set is fixed and a seed's draw is unavailable if any checkpoint leaves the reference's range, and a seed with more than 10% unavailable draws makes the contrast "not interpreted (support)". Curve raised: pooled gap $\geq +2$, 95% interval above 0, every seed positive, Holm $p \leq 0.05$. Average equivalence: pooled (and, from Pre-registration 20, each seed's) 90% interval within ± 2. Arm beats arm (method contrasts, one arm's checkpoints against the other's trajectory): pooled $\geq +0.5$, 95% interval above 0, every seed positive, Holm $p \leq 0.05$. Intervals condition on the trained seeds.

Table 1. Registered matched-pass@1 differences in pass@16 (points; 95% intervals unless marked 90%). Reference: Dr. GRPO (B) trajectory or trajectory-and-temperature envelope. 1.5B unless stated.

Comparison	Setting	Difference
NSR vs PSR	512 tok., 3 seeds	−0.20 [−0.97, +0.55] (90%)
Coverage-weighted replay vs B	512, 3 seeds	+1.19 [+0.54, +1.79]
Allocation (frontier value) vs B	1,024, 3 seeds	−0.12 [−0.60, +0.29]
PSR / NSR vs B	1,024, 3 seeds	+0.11 / −0.15
Human / self prefixes vs B	1,024, 3 seeds	−0.05 / −0.11
UCB-Con bonus vs B	1,024, 3 seeds	+0.40 [+0.02, +0.76]
Batch bonus vs B	1,024, 3 seeds	+0.01 [−0.38, +0.39]
Flat-RLVR (ρ 10 / 5) vs B	1,024, 3 seeds	+0.29 [−0.09, +0.69] / −0.02
PSR / NSR / allocation vs B	7B, 2,048, 2 seeds	−0.09 / −0.09 / −0.15 (90%)

3. RL-side interventions stay on one curve

Table 1 collects registered comparisons. Advantage rules (PSR, NSR, constant baselines, noise injection, pass@k objectives, MaxRL), entropy terms, replay and group completion, larger groups (32 rollouts), rollout allocation by frontier value and its controls, human-solution and self-generated prefix guidance, process credit, update-geometry changes and sharpness-aware updates either reach average equivalence or stay within ± 1.2 points with intervals including zero; full verdicts are in Appendix A. The UCB-Con bonus raises distinct valid incorrect answers per problem by 0.057 (its registered diversity rule) without moving the curve (+0.40); its matched pass@32 gap (+0.31, 95% $[-0.25, +0.78]$) excludes the +1.7-point magnitude reported by Song et al. (2025), whose training, model and implementation differ. Allocation by frontier value raises the share of success-bearing groups among pilot-zero problems from 18–21% to 50–54% and still stays on the curve (forecast +4.1, observed −0.05): lack of on-policy signal on rare problems is not, in this form, what fixes the curve.

Longer training. Dr. GRPO for 1,000 steps at learning

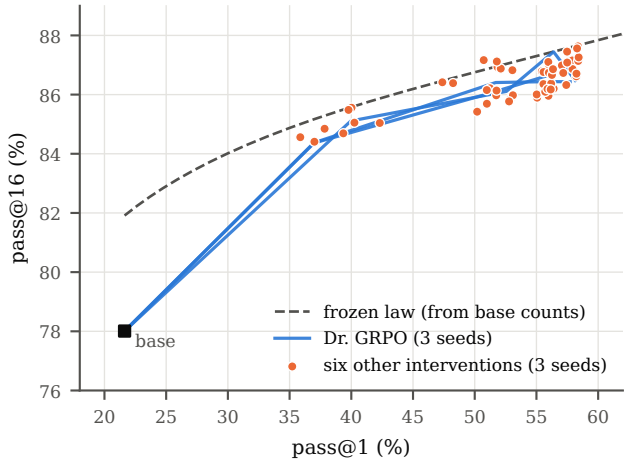


Figure 1. 1.5B at 1,024 tokens: Dr. GRPO trajectories (three seeds) and checkpoints of six interventions (three seeds each). Dashed: a frozen per-problem law computed from the base model's success counts (Appendix B).

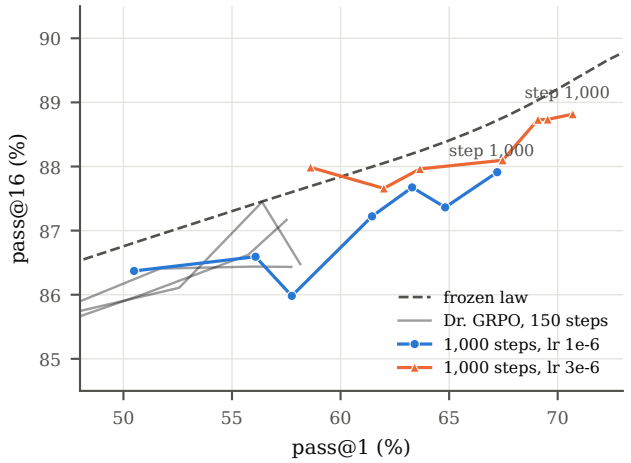


Figure 2. Dr. GRPO for 1,000 steps at two learning rates against the 150-step trajectories and the frozen law.

rates 10^{-6} and 3×10^{-6} raises pass@1 from 57.8 to 67.2 and from 63.7 to 70.7; pass@256 at step 1,000 minus the base is +1.96 [+0.39, +3.72] and +0.39 [−1.37, +2.15], with Holm-adjusted one-sided $p = 0.029$ and 0.358 against the registered 0.025: not supported, the first borderline (Figure 2). The base model's difficulty order is largely preserved (Spearman 0.67–0.72).

4. Public checkpoints

To test whether the curve is specific to our recipe, we evaluated 28 published checkpoints from the same base models under one protocol (our prompt, $T = 1$, 32 samples, pinned revisions): 14 at 1.5B (1,024 and 3,072 tokens) and 14 at 7B (2,048 tokens), spanning Dr. GRPO, GRPO, DAPO and RAFT++ recipes, one-shot RLVR, rank-1 extrapolation, replay (Zhan et al., 2025), off-policy guidance (Yan et al., 2025), interleaved SFT (Ma et al., 2025) and random-reward reproductions (Shao et al., 2025). Two truncate more than 90% of answers; the remaining 26 give 38 comparisons against the frozen law at the same budget (Figure 3). Residuals range from −7.6 to +0.9 points; 11 are equivalent, 22 inconclusive, 5 depart downward (mostly truncation-heavy), none upward. A registered sensitivity analysis at 7B with a dense empirical reference (28 comparisons) gives 6 equivalent, 18 inconclusive and 4 downward departures, none upward. A common protocol is a control and a limitation: checkpoints may do better under their native prompts and budgets.

5. Warm starts lift the curve during early RL

If RL cannot change the curve, can the starting model? Each start is a small SFT before unchanged Dr. GRPO (nominal dose 25K presented response tokens, AdamW 10^{-5}, batches of 8; dose chosen on validation after larger doses moved pass@1 beyond the reference's range). Table 2 lists the registered verdicts.

Starts. H25: verified solutions of a stronger model (Qwen2.5-Math-7B-Instruct) to 262 training problems with at most 4 of 64 base successes; R25: the same number of random teacher-covered problems. TEACHF and SELF: on 163 hard problems, 256 length-binned pairs of a teacher solution and a verified, finished base-model solution of the same problem (mean 470 and 466 tokens), so the two differ only in the solution's source. CORR: H25's teacher solutions with the final boxed answer replaced by a wrong one. FMT: base responses with the loss only on box delimiters (its manipulation check failed: boxing did not increase). CWS, UNI, HARD, TAIL: the base model's own verified solutions on 3,795 problems, drawn with weight $16(1-\hat{p}_x)^{15}$ (the marginal pass@16 value of Proposition 2.1), uniformly, only on $c \le 4$ problems, or uniformly with TailSFT's token filter (Malladi et al., 2026); TCWS: coverage weights on teacher solutions; TIME: SELF's stream interleaved during RL (one SFT batch every five steps); OPD: on-policy distillation from the teacher at the same dose (Dong et al., 2026).

Teacher content, not labels, timing or weighting. The cleanest contrast holds prompts and lengths fixed and changes only who wrote the solutions: teacher solutions beat the base model's own (+1.53, all seeds positive). Corrupting the final answer leaves the effect unchanged (CORR minus H25 +0.22, 90% $[-0.34, +0.82]$), so the carrier is the reasoning, not the answer key; a second SFT seed reproduces H25

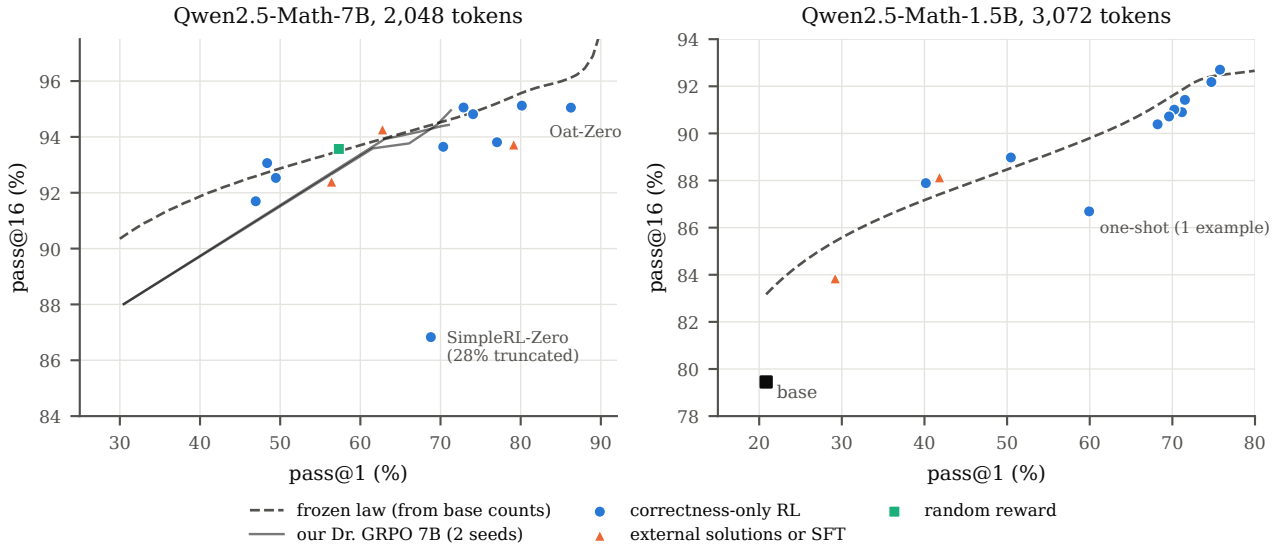


Figure 3. Public checkpoints against the frozen law computed from the base model at the same budget. Left: 7B at 2,048 tokens. Right: 1.5B at 3,072 tokens.

Table 2. Registered warm-start verdicts (1.5B, 1,024 tokens, three RL seeds, RL steps 1–150 where within the reference's range). Gap: pass@16 at matched pass@1 (points, 95% interval). Against Dr. GRPO unless stated.

Start (vs reference)	Gap	Verdict
H25 (teacher, hard)	+1.92 [+0.68, +3.12]	inconclusive (< 2)
TEACHF (teacher, matched prompts)	+2.10 [+0.86, +3.47]	curve raised
CORR (teacher, wrong final box)	+2.14 [+1.04, +3.35]	curve raised
H25b (H25, second SFT seed)	+1.86 [+0.71, +3.14]	inconclusive (< 2)
SELF (base's own, matched prompts)	+1.78 [+0.86, +2.70]	inconclusive (< 2)
TEACHF vs SELF trajectory	+1.53 [+0.34, +2.76]	teacher beats self
CWS vs UNI / HARD / TAIL / SELF	+0.03 / −0.07 / +0.47 / −0.31	none beaten
SELF before RL vs TIME (during RL)	+1.02 [+0.29, +1.71]	before better (pre-dose)
after TIME's dose completes	−0.03 [−0.62, +0.57]	descriptive
OPD vs R25 trajectory	−1.02 [−2.25, +0.21]	inconclusive

(−0.06). Among self-generated starts, coverage weighting, hard-only selection and TailSFT are indistinguishable (all CWS contrasts within ±0.5; 90% intervals within ±1, a descriptive, unadjusted equivalence), and interleaving the same data during RL is equivalent to applying it before RL once the dose is complete. Coverage-weighted teacher solutions beat coverage-weighted self solutions by +1.63 [+0.62, +2.62] (not interpreted: support in one seed). On-policy distillation at this dose gives no gain over plain SFT.

A frozen law does not forecast starts. A per-problem law fitted on Dr. GRPO and frozen (Appendix B) forecast +6.69 for H25; the observed +1.92 gives a residual of −4.77 [−6.57, −2.88]: registered verdict "forecast fails". The law describes RL from a given start, not how a start changes the curve.

Budget. At 1,024 tokens the starts truncate 45–48% of answers. At 2,048 tokens (three seeds), H25's early-RL gap is +1.70 [+0.50, +2.88] and its start offset +2.5 to +3.3; over checkpoints 50–150, TEACHF keeps +1.72 [+0.61, +2.83] while SELF (−0.03), CWS (+0.27) and UNI (−0.08) are near zero (descriptive).

6. The excursion is transient

Teacher starts reach pass@1 63–65 by step 150, beyond the 150-step reference's range (about 58), so their late checkpoints could not be read in Table 2. Three 700-step Dr. GRPO replicates (seeds 8–10) reproduce the 150-step references within 2 points of pass@1 and reach pass@1 65.2–65.5 and pass@16 88.4–89.0 at step 700. The registered extension of the H25 and R25 contrasts to these references is not interpretable (support fails in two seeds; H25's estimate is +1.54). Descriptively (Figure 4), every start's gap decays. Averaged over seeds, TEACHF is +2.1 at steps 1–10, +1.3 at step 50, +1.2 at step 100 and +0.5 at step 150; CORR +2.3, +1.8, +0.3, +0.4; H25 +1.8, +1.9, +0.4, +0.4; SELF +2.8, +0.6, +0.6, +0.5. Self starts decay by step 50, teacher starts by steps 100–150. Teacher starts end where Dr. GRPO arrives after about 700 steps, with the same pass@16: they move faster along the frontier and leave it only transiently. The registered P36 secondary reading against these references (six arms, Holm) raises no curve; all estimates are +0.7 to +1.6.

7. Where the gains sit

Spread, not answer variety. Across ten RL-side conditions, matched pass@2 gaps are +0.08 to +0.19 against +0.13 between two Dr. GRPO seeds; H25's is +2.39 [+1.97, +2.82] (90%) and self-generated starts'

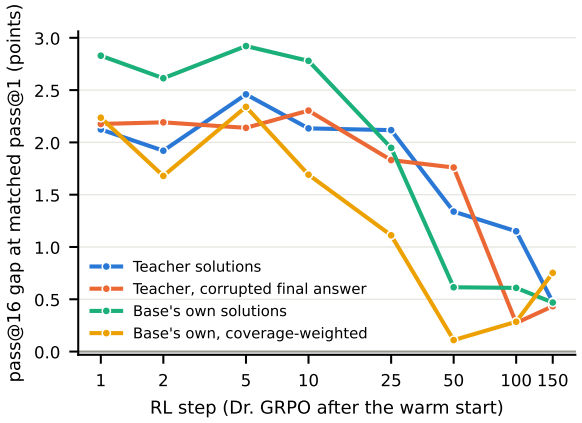


Figure 4. Matched-pass@1 pass@16 gap by RL step after four warm starts, against 700-step Dr. GRPO replicates (mean of seeds 8–10; descriptive). Points lie inside the replicate's pass@1 range in all three seeds.

about +1.2. By Proposition 2.1, the coverage gains are a more even distribution of success across problems. Within problems, the two kinds of start differ: self starts increase distinct incorrect answers (+1.3 per problem) while the teacher start reduces them (−1.5); both lower the chance that two incorrect answers coincide. During Dr. GRPO itself (seed 8), that chance rises from 5% to 14% while the across-problem variance of success rises from 0.033 to 0.130.

Which problems. An exact decomposition of the matched gap by base-model success count (the reference is interpolated at overall pass@1 and the same weights are applied within each stratum) shows that self starts' early gain sits on problems with 1–8 of 64 base successes (+1.4 to +1.7 of +1.8 to +2.0 points at steps 1–10) and is gone by steps 50–150, whereas H25's gain moves toward problems with no base success (+0.86 at steps 1–10, +1.52 of +2.01 at step 50). Self-generated starts reweight successes the base can already sample, which RL recovers on its own; teacher reasoning supplies successes on problems the base does not sample, which RL takes longer to match.

Finishing. Part of RL's own gain is learning to finish an answer: at 1,024 tokens the base leaves 42.5% of its answers without a final box; forcing a final answer on the base recovers 16% of Dr. GRPO's pass@1 gain (Appendix C).

8. Related work

Yue et al. (2025) find base models beating RLVR models at large k; the invisible-leash analysis argues RLVR mostly reweights within the base model's support (Wu et al., 2025), and RL is reported to squeeze while SFT expands reasoning trajectories (Matsutani et al., 2025). We test these claims at matched accuracy with registered rules, across interventions, public checkpoints and warm starts. Diversity-oriented RL methods include pass@k objectives (Walder & Karkhanis, 2025; Chen et al., 2025), outcome-based exploration (Song et al., 2025), entropy control (Cui et al., 2025), rollout scaling and allocation (Hu et al., 2025; Yang et al., 2025; Li et al., 2025b; Yu et al., 2026) and sharpness-aware updates (Kim et al., 2026); most report fixed-step comparisons, and those we reproduced stay on the curve. External guidance during RL (Yan et al., 2025; Ma et al., 2025; Li et al., 2025a; Zhang et al., 2025b; Ding et al., 2026) and self-generated rationalizations (Zelikman et al., 2022; Zhou et al., 2025; Mahdavi et al., 2026) inject information the policy lacks; our warm-start results isolate the content of that information at a fixed dose. Cold-start SFT before RL is standard (Guo et al., 2025); on-policy distillation before RL (Dong et al., 2026), TailSFT (Malladi et al., 2026), mid-training (Wang et al., 2025; Zhang et al., 2025a) and distillation (Kim et al., 2025) change what RL can reach; difficulty-aware rejection tuning (Tong et al., 2024) allocates synthesis to hard problems. Budgeted-search views map RL operating points to base sampling budgets (Sun et al., 2026).

9. Discussion

Within the tested settings, RLVR algorithms set how fast a model moves along a coverage-versus-accuracy frontier, the base model sets the frontier, and warm starts change the position and speed along it, with a transient excursion whose size depends on whether the start carries reasoning the base cannot sample. A method that durably raises coverage at matched accuracy would have to change what the policy can reach, not how RL weights what it already samples. Two open questions are being tested under registered rules [Pre-registration 39 running]: whether a $k = 16$ instrument near ceiling hides RL-side effects at pass@128 on harder held-out problems (OlympiadBench and AIME, deduplicated against MATH), and whether verified solutions harvested by deep sampling of the base model on problems it never solves in 64 attempts behave like teacher content.

10. Limitations

Models are Qwen2.5-Math 1.5B and 7B on MATH-style problems with binary verification; answer budgets of 512–3,072 tokens are part of the estimand. Bootstrap intervals cover test-problem sampling con-

ditional on the trained seeds. Average equivalence within ±2 points permits smaller effects; several comparisons are inconclusive or lack reference support rather than being equivalent. Warm-start arms ran on A100s and the Dr. GRPO references on L40S; within-hardware contrasts (TEACHF vs SELF, CWS vs baselines) are unaffected. The persistence analysis uses three 700-step replicates and is descriptive. pass@16 near 87–89 limits headroom; the base-zero test stratum has 51 problems. A second model family is being prepared.

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A. Registrations and verdicts

Reg.	Comparison	Verdict
9–11	NSR vs PSR (OLMo, Qwen); noise injection; temperature	Average equivalence; temperature sweep -1.42 (90% $[-2.19, -0.68]$)
12, 14	Coverage-weighted and base-seeded replay; larger groups (G32)	$+1.19$, $+1.02$: below $+2$; G32 exploratory, beyond range
16	Answer budget (check)	Warm-start gain $+4.77 \to +2.15$ from 512 to 1,024 tokens: largely a budget effect
17	Frozen per-problem law	RMSE criterion met on three test sets; fails after a warm start
18	Rollout allocation (frontier value, $\mathbb{E}[p(1-p)]$, HORA)	No rule met; forecast not confirmed
19, 22	Human-solution warm starts; human vs self prefixes	Not interpretable (no overlap); prefixes on the curve
20, 21	7B at 2,048 tokens; 1.5B at 1,024 tokens	3 of 4 and 4 of 4 contrasts equivalent
23, 26	Minimum-help credit; update-direction, whitening and teacher-gradient pilots	None selected
27	1,000-step training	Not supported (LH1 borderline)
28, 33	Public checkpoints; 7B sensitivity	No upward departure
30	Teacher warm starts (dose screen; H25, R25; frozen-law forecast)	H25 inconclusive ($+1.92$); R25 not interpreted; forecast fails
31	Flat-RLVR	Average equivalence (both radii)
32	UCB-Con and Batch outcome bonuses	Average equivalence; UCB-Con raises distinct incorrect answers
34	On-policy distillation start	Primary not interpreted; secondary inconclusive (-1.02)
35, 36	SELF/TEACHF/CORR/FMT/H25b; CWS, UNI, HARD, TAIL, TCWS, TIME	Table 2
39	Hard held-out sets, pass@128 re-evaluation, harvest pilot, retention	[running]

B. The frozen per-problem law

Each problem's log-odds start at an empirical-Bayes estimate from the base model's 32-sample counts and follow $\dot{\theta}_x = p_x(1-p_x) + \tau \frac{1}{N} \sum_{x'} p_{x'}(1-p_{x'})$ with $\tau = -0.20$, fitted once on Dr. GRPO's 1.5B checkpoints and frozen; a checkpoint is placed where the law's mean success equals its observed pass@1, and pass@16 is then a prediction. It describes the curve conditional on pass@1, not training speed.

C. Finishing

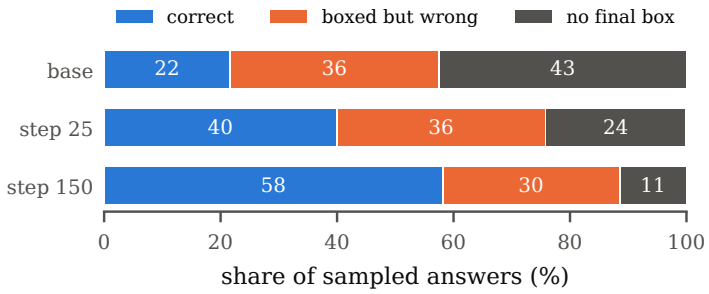


Figure 5. Answers of the 1.5B base and Dr. GRPO (seed 8) at steps 25 and 150 at 1,024 tokens: correct, boxed but wrong, and without a final box.

Between the base and step 25, correct answers rise by 18.4 points while unfinished answers fall by 18.3. Appending "Final answer: \boxed{" to unfinished base samples and decoding 32 greedy tokens raises pass@1 from 22.0 to 27.7 (22.3 with the same extra tokens and no cue).

D. Long references

The 700-step replicates B1kL8–10 (L40S, the B1k recipe) reach pass@1/pass@16 of 39.2/84.9, 51.7/85.7, 56.4/86.3, 58.3/86.7 (seed 8) at steps 25, 50, 100, 150, against 40.0/85.1, 52.6/86.1, 56.4/87.4, 58.1/86.5 for B1k8; at steps 200–700 seed 8 reaches 59.7/87.6, 60.9/87.8, 61.8/87.9, 63.2/88.1, 63.7/87.7, 65.5/88.5.